Application of Statistically-Based Experimental Designs for the Optimization of Eicosapentaenoic Acid Production by the Diatom *Nitzschia laevis*

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Abstract: Statistically based experimental designs were applied to the optimization of medium components and environmental factors for eicosapentaenoic acid (EPA) production by the diatom *Nitzschia laevis* in heterotrophic conditions. First, the Plackett-Burman design was used to evaluate the effects of variables including medium components and environmental factors on cell growth and EPA production. Among these variables, NaCl, $CaCl_2$, PI metal solution, pH, and temperature were identified to have the significant effects (with confidence level > 90 %). Subsequently, the concentrations of NaCl, $CaCl_2$, PI metal solution as well as the values of pH and temperature were optimized using central composite design. The cell growth and EPA production were found to respectively correlate to NaCl, $CaCl_2$, pH, and temperature that could be represented by second-order polynomial models. The optimal values of the four parameters were determined by response surface and numerical analyses as 8 g/L NaCl, 0.10 g/L $CaCl_2$, pH 8.5 and 19.8°C for cell dry weight (DW), and 14 g/L NaCl, 0.10 g/L $CaCl_2$, pH 8.5 and 18°C for EPA production, respectively. The subsequent verification experiments confirmed the validity of the models. This optimization strategy led to a DW of 9 g/L, an EPA yield of 280 mg/L and an EPA productivity of 28 mg/L/d, respectively, which were considerably higher than those obtained in the previous studies. © 2001 John Wiley & Sons, Inc. *Biotechnol Bioeng* **75**: 159–169, 2001.

Keywords: diatom; *Nitzschia laevis;* EPA; heterotrophic; optimization; experimental design

INTRODUCTION

The therapeutic significance of ω-3 polyunsaturated fatty acids (ω-3 PUFAs) has been demonstrated by recent clinical studies. Eicosapentaenoic acid (EPA), along with docosahexaenoic acid (DHA), plays an important role in the prevention of arrhythmia, cardiovascular disease, and cancer (Gill and Valivety, 1997). Currently, fish oil is the commercial source of EPA and DHA. However, the production of ω-3 PUFAs from fish oil is hindered by limitations such as stability problems, high purification cost and contamination with pesticides and heavy metals. As fish obtain ω-3 PUFAs from zooplankton that consume microalgae, many efforts have been devoted to producing ω-3 PUFAs directly from microalgae (Barclay et al., 1994). To date, there have been some reports about the EPA recovery from some microalgae such as *Monodus subterraneus* and *Phaeodactylum tricornutum* (Belarbi et al., 2000). The authors showed that the quality of microalgal EPA is superior to that of fish oil product and the accepted cost is competitive to fish oil.

Several attempts have been made to develop commercial heterotrophic processes for ω-3 PUFAs production because a conventional photoautotrophic process is disadvantageous due to light limitation resulting from cells' mutual shading (Chen, 1996; Vazhappilly and Chen, 1998). Recently, the diatom, *Nitzschia laevis*, was studied in our laboratory and the results showed that it is a good EPA producer as the cells showed not only good growth but a relatively high EPA content under heterotrophic conditions (Wen and Chen, 2000). To maximize EPA production by this alga, determining the optimal culture conditions (including medium components and environmental factors) is needed. We had investigated the effects of silicate, glucose, and nitrogen source on EPA production by *N. laevis*. Although good results were obtained, information about the influence of other medium components (e.g., salts, trace elements, vitamin B_{12}, biotin) and environmental factors (e.g., pH, temperature) is still insufficient. Culture conditions are traditionally optimized by the one-at-a-time strategy, i.e., varying one factor while keeping all others constant. Although the strategy is simple and easy, without the need of statistical analysis, it involves a relatively large number of experiments and the interaction among factors is often ignored. In contrast, statistically based experimental designs

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are more efficient approaches that can deal with a large number of variables simultaneously. Moreover, the interaction among different variables can be estimated. To date, this technique has been used by some researchers to optimize medium components (Gong and Chen, 1997; Ooijkaas et al., 1999; Rao et al., 2000) and/or environmental factors as well as operation parameters (Gobbetti et al., 1999; Kalil et al., 2000; Lay, 2000; Souza et al., 1999) in various fermentation processes.

The aim of this work was to optimize the heterotrophic production of EPA by the diatom *N. laevis* using statistically based experimental designs. In the first step, the Plackett-Burman design was used to determine the likely effects of medium components (i.e., salts, trace metal elements, vitamin B_{12}, biotin) and environmental factors (i.e., pH, temperature) on cell growth and EPA production. Subsequently, the concentrations of the medium components and levels of the environmental factors that had significant effects were optimized using a central composite design and response surface analyses.

MATERIALS AND METHODS

Cell Line, Medium, and Culture Conditions

The diatom *Nitzschia laevis* (UTEX 2047) was used in this work. The cells were subcultured in LDM medium supplemented with 5 g/L glucose and 30 mg/L $Na_2SiO_3 \cdot 9H_2O$. The LDM medium consisted of (per L) 1 g tryptone, 892 mL artificial seawater containing salts, tris-buffer and chelated iron (Starr and Zeikus, 1993), 100 mL bristol solution including some salts (Starr and Zeikus, 1993), 6 mL PIV metal solution including Na_2EDTA and trace elements (Starr and Zeikus, 1993), 1 mL stock solutions of biotin (25.0×10^{-5} g/L) and vitamin B_{12} (15.0×10^{-5} g/L), respectively. The cells were incubated in 500-mL Erlenmeyer flasks containing 200 mL medium in an orbital shaker (at 160 rpm) at 22°C in darkness. In medium preparation, if initial pH is higher than ≈ 8.5, precipitation would occur after autoclaving. So, the following steps were taken to avoid this problem. (1) Medium in each flask was first adjusted to pH 7.5 before autoclaving; at this pH level, no precipitation occurred even at high calcium and trace element concentrations. (2) After autoclaved at 121°C for 20 min, the flasks were cooled down to room temperature. (3) For adjustment of each desired pH level, one flask was taken out and the added amounts of acid or base were recorded. (4) The same amounts of acid or base were added to the parallel identical flasks; it was found the final pH of these flasks was rather accurate. By above operation, no precipitation occurred.

The cell concentration of the inoculum was about 0.1 g/L (DW). Our previous data indicated that EPA production of *N. laevis* was improved by an increase in initial glucose and silicate concentrations, and 20 g/L glucose and 120 mg/L $Na_2SiO_3 \cdot 9H_2O$ were found to be the most appropriate. Also, addition of 1.6 g/L tryptone and 0.8 g/L yeast extract could enhance EPA production by the alga (data not shown). So these concentrations were used in the entire experiments (i.e., screening, optimization, and verification). All the experiments were repeated.

Determination of Cell Dry Weight and Fatty Acid Composition

Cell dry weight (DW) was determined by drying as reported previously (Wen and Chen, 2000). The cells at early stationary phase were harvested and lyophilized for fatty acids analysis. Fatty acid methyl esters were prepared by transmethylation with methanol-acetyl chloride (Cohen et at, 1997) and analyzed by GC. The detailed GC analytical procedures were previously described by Wen and Chen (2000).

Experimental Designs and Data Analyses

Plackett-Burman Design

The first optimization step was to identify which variables have significant effects on EPA production by *N. laevis*. The variables to be evaluated are listed in Table I, that include some medium components (i.e., salts, trace metal elements, vitamin B_{12}, biotin) and environmental factors (i.e., pH, temperature). The Plackett-Burman (PB) design is very useful in screening for the most important factors from a lot of candidates (Lee et al., 1999). As shown in Table II, the design matrix was developed according to Montgomery (1997), there are 13 variables (A–M) over 20 runs in the design; D_1 to D_6 are dummy variables employed to evaluate the standard errors of the experiment. Each independent variable was investigated at a high (+) and a low (−) level. The low levels (−) of medium components were taken as their concentrations in LDM medium, while pH and temperature were taken as those in subculture conditions (Table

Table I. Variables to be screened in Plackett-Burman design and real values for the two levels of the variables.[a]

	Variables	Unit	Low level (−)	High level (+)
A	NaCl	g/L	16	32
B	$MgSO_4 \cdot 7H_2O$	g/L	2.2	4.4
C	KCl	g/L	0.54	1.08
D	$CaCl_2$	g/L	0.204	0.408
E	KH_2PO_4	g/L	0.04	0.11
F	Tris buffer	g/L	0.9	1.8
G	Vitamin B12 solution	mL/L	1.9	3.8
H	Biotin solution	mL/L	1.0	2.0
I	PI metal solution	mL/L	8.92	17.84
J	Chelated iron solution	mL/L	2.68	5.36
K	PIV metal solution	mL/L	6	12
L	Initial medium pH	—	7.5	9.0
M	Temperature	°C	22	30

[a]The components of PI, PIV, and chelated solutions are ions including Fe^{3+}, Mn^{2+}, Co^{2+}, Zn^{2+}, and Mo^{6+} (Starr and Zeikus, 1993).

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Table II. Placket-Burman design of variables (in coded levels) with DW, EPA content, and EPA yield as responses.

	Variables																			Responses		
Run	A	B	C	D	E	F	G	H	I	J	K	L	M	D_1	D_2	D_3	D_4	D_5	D_6	DW (g/L)	EPA content (% DW)	EPA yield (mg/L)
1	+	+	−	−	+	+	+	+	−	+	−	+	−	−	−	−	+	+	−	6.86	2.73	187.278
2	−	+	+	−	−	+	+	+	+	−	+	−	+	−	−	−	−	+	+	0.28	2.56	7.168
3	+	−	+	+	−	−	+	+	+	+	−	+	−	+	−	−	−	−	+	6.64	2.84	188.576
4	+	+	−	+	+	−	−	+	+	+	+	−	+	−	+	−	−	−	−	0.28	2.46	6.888
5	−	+	+	−	+	+	−	−	+	+	+	+	−	+	−	+	−	−	−	7.62	2.81	214.122
6	−	−	+	+	−	+	+	−	−	+	+	+	+	−	+	−	+	−	−	0.32	2.44	7.808
7	−	−	−	+	+	−	+	+	−	−	+	+	+	+	−	+	−	+	−	0.28	2.23	6.244
8	−	−	−	−	+	+	−	+	+	−	−	+	+	+	+	−	+	−	+	0.32	2.35	7.520
9	+	−	−	−	−	+	+	−	+	+	−	−	+	+	+	+	−	+	−	0.32	2.39	7.648
10	−	+	−	−	−	−	+	+	−	+	+	−	−	+	+	+	+	−	+	8.08	2.82	227.856
11	+	−	+	−	−	−	−	+	+	−	+	+	−	−	+	+	+	+	−	6.66	2.75	183.150
12	−	+	−	+	−	−	−	−	+	+	−	+	+	−	−	+	+	+	+	0.40	2.83	11.320
13	+	−	+	−	+	−	−	−	−	+	+	−	+	+	−	−	+	+	+	0.24	2.43	5.832
14	+	+	−	+	−	+	−	−	−	−	+	+	−	+	+	−	−	+	+	6.86	2.87	196.882
15	+	+	+	−	+	−	+	−	−	−	−	+	+	−	+	+	−	−	+	0.32	2.44	7.808
16	+	+	+	+	−	+	−	+	−	−	−	−	+	+	−	+	+	−	−	0.32	2.46	7.872
17	−	+	+	+	+	−	+	−	+	−	−	−	−	+	+	−	+	+	−	7.42	2.81	208.502
18	−	−	+	+	+	+	−	+	−	+	−	−	−	−	+	+	−	+	+	7.52	2.91	218.832
19	+	−	−	+	+	+	+	−	+	−	+	−	−	−	−	+	+	−	+	6.18	2.79	172.422
20	−	−	−	−	−	−	−	−	−	−	−	−	−	−	−	−	−	−	−	7.96	2.76	219.696

I). A high level (+) of each factor is far enough apart from the low level so that a significant effect, if it exists, is likely to be detected.

Statistical analyses were used to identify the variables that had significant effects on the responses (i.e., DW, EPA content, and EPA yield). The effect of each variable (E_i) on each response was determined by subtracting the average response of the low level from that of the high level. Once the dummy variables are concerned, their effects are used to calculate the standard error (*SE*) as follows:

$$SE = \sqrt{\frac{\Sigma(E_d)^2}{n}} \tag{1}$$

where E_d is the effect of each dummy variable and n is the number of dummy variables. The significance of each variable can be determined using Student's *t*-test as follows,

$$t - value = E_i / SE \tag{2}$$

Only confidence levels above 90% were accepted in this experiment. Accordingly, it was possible to rank the variables with respect to their effects on DW, EPA content, and EPA yield, respectively.

Central Composite Design

Once the variables having the greatest influence on the responses were identified, a central composite design (CCD) was used to optimize the levels of these variables (Montgomery, 1997). As shown in Table III, the design matrix is a 2^{5-1} fractional factorial design combined with 6 central points (run 27–32) and 10 axial points where one variable is set at an extreme level (±2) while the other variables are set at their central point level. The true values of the variables are given in Table IV. Coding of variable *i* in the CCD was done as follows:

$$x_i = (X_i - X_{cp})/\Delta X_i \qquad i = 1, 2, 3, \ldots, k \tag{3}$$

where x_i is the coded level, X_i is the real value, X_{cp} is the real value at central point, and ΔX_i is the step change of variable *i*, respectively.

The cell dry weight and EPA production can be written as the functions of the independent variables by second-polynomial, i.e.,

$$Y = \beta_0 + \Sigma\beta_i x_i + \Sigma\beta_{ii} x_i^2 + \Sigma\beta_{ij} x_i x_j \tag{4}$$

where Y is the predicted response, β are the coefficients of the equation, and x_i and x_j are the coded levels of variables *i* and *j*, respectively. This equation can be used to evaluate the linear, quadratic, and interactive effects of independent variables on the chosen response.

After the responses for each run were obtained (Table III), they were subjected to multiple nonlinear regression using the software STATISTICA (V5.0, StatSoft Inc., OK, 1995) to obtain the coefficients of the second polynomial [Eq. (4)]. Only the estimates of coefficients with significant levels higher than 90% (i.e., $p < 0.10$) were included in the final models. The *F*-test was used to evaluate the significance of the models.

RESULTS AND DISCUSSION

Screening Experiments

Initial experiments were carried out according to the PB design matrix as shown in Table II. The cells were cultured

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Table III. Central composition design of variables (in coded levels) with DW, EPA content, yield, and productivity as responses.[a,b]

	Variables					Responses			
Run	NaCl	$CaCl_2$	PI	pH	T	DW (g/L)	EPA content (% of DW)	EPA yield (mg/L)	EPA productivity (mg/L/d)
1	−1	−1	−1	−1	+1	4.90	3.00	147.0	14.70
2	−1	−1	−1	+1	−1	7.74	3.53	273.2	27.32
3	−1	−1	+1	−1	−1	7.23	3.38	244.4	22.22
4	−1	−1	+1	+1	+1	6.13	2.88	176.5	18.58
5	−1	+1	−1	−1	−1	6.84	3.56	243.5	21.17
6	−1	+1	−1	+1	+1	7.03	2.89	203.1	22.57
7	−1	+1	+1	−1	+1	5.57	3.05	169.9	16.99
8	−1	+1	+1	+1	−1	6.55	3.78	247.6	21.53
9	+1	−1	−1	−1	−1	7.05	3.59	253.1	21.09
10	+1	−1	−1	+1	+1	5.80	2.83	164.1	18.24
11	+1	−1	+1	−1	+1	4.32	2.96	127.9	12.79
12	+1	−1	+1	+1	−1	7.40	3.38	250.1	22.74
13	+1	+1	−1	−1	+1	4.25	2.94	124.9	12.50
14	+1	+1	−1	+1	−1	6.40	3.68	235.5	20.48
15	+1	+1	+1	−1	−1	5.11	3.86	197.2	16.44
16	+1	+1	+1	+1	+1	5.75	2.90	166.7	18.53
17	−2	0	0	0	0	8.00	2.37	189.6	18.06
18	+2	0	0	0	0	7.70	2.20	170.9	17.09
19	0	−2	0	0	0	7.48	2.67	199.7	18.16
20	0	+2	0	0	0	7.34	3.04	233.1	21.25
21	0	0	−2	0	0	8.06	2.68	216.0	22.74
22	0	0	+2	0	0	7.51	2.91	218.5	20.81
23	0	0	0	−2	0	7.49	2.98	223.2	18.60
24	0	0	0	+2	0	7.97	2.88	229.5	20.87
25	0	0	0	0	−2	6.15	3.97	244.2	18.78
26	0	0	0	0	+2	1.60	2.32	37.12	5.30
27–32	0	0	0	0	0	7.69	2.97	228.8	22.87

[a]The presented data of responses in run 27–32 are means of six independent replicates.
[b]The EPA productivity was determined by calculating the EPA yield over the culture time.

Table IV. Real values of independent variables in central composition design.

Level	NaCl (g/L)	$CaCl_2$ (g/L)	PI solution mL/L	pH	Temperature (°C)
+2	24	0.309	13.50	8.5	28
+1	20	0.257	11.25	8.0	25
0	16	0.204	9.00	7.5	22
−1	12	0.151	6.75	7.0	19
−2	8	0.098	4.50	6.5	16

under 20 different culture conditions. DW, EPA content, and EPA yield were taken as the responses for each run (Table II). Table V shows the resulting effects of the variables on the responses, and the associated *t*-values and significant levels. Confidence levels were accepted only when above 90%. The significance of dummy variables (D_1–D_6) were never as high as 90% (data not shown). As shown in Table V, NaCl, $CaCl_2$, the PI metal solution, pH, and temperature (T) in the tested range had significant effects on

Table V. The effects of variables in PB design on DW, EPA content, and EPA yield and associated statistical tests.

	DW			EPA content			EPA yield		
Variables	Effects	*t*-Value	*p*-Level	Effects	*t*-Value	*p*-Level	Effects	*t*-Value	*p*-Level
NaCl	−0.552	−4.701	**0.003**	−0.036	−0.688	0.517	−16.47	−4.184	**0.006**
$MgSO_4 \cdot 7H_2O$	0.200	1.703	0.139	0.090	1.720	0.136	5.797	1.472	0.191
KCl	−0.020	−0.170	0.870	0.022	0.420	0.689	0.592	0.150	0.886
$CaCl_2$	−0.244	−2.078	**0.083**	0.060	1.146	0.295	−4.273	−1.085	0.320
KH_2PO_4	−0.080	−0.681	0.521	−0.076	−1.452	0.197	−2.253	−0.572	0.588
Tris buffer	−0.168	−1.431	0.202	−0.006	−0.115	0.912	−3.832	−0.973	0.368
Vitamin B12 solution	−0.148	−1.260	0.254	−0.058	−1.108	0.310	−5.080	−1.290	0.244
Biotin solution	−0.040	−0.341	0.745	−0.046	−0.879	0.413	−1.066	−0.271	0.795
PI metal solution	−0.264	−2.248	**0.066**	0.050	0.955	0.376	−7.879	−2.001	**0.092**
Chelated iron solution	0.168	1.431	0.202	0.064	1.223	0.267	5.889	1.496	0.185
PIV metal solution	−0.128	−1.090	0.318	−0.036	−0.688	0.517	−3.668	−0.932	0.385
Initial medium pH	−0.232	−1.976	**0.095**	−0.010	−0.191	0.855	−7.201	−1.829	0.117
Temperature	−6.872	−58.52	**0.000**	−0.350	−6.688	**0.001**	−194.1	−49.31	**0.000**

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DW, while only T significantly influenced the EPA content. For EPA yield, the significant variables were NaCl, PI metal solution, and T. Based on these results, the factors having the greatest impacts on the production of EPA by *N. laevis* were identified as NaCl, $CaCl_2$, the PI metal solution, pH, and temperature.

The concentration of NaCl (salinity) is an important factor influencing the growth and chemical composition of microalgae. It has been reported that the growth and DHA content of the marine microalga *Crypthecodinium cohnii* decreased when salinity was doubled from 15 g/L to 30 g/L NaCl (Jiang and Chen, 1999). Also, the red alga *Porphyridium cruentum* produced less polyunsaturated fatty acids when growth was retarded by increased salinity (Cohen et al., 1988). In the present work, the negative effect of NaCl in the concentration range tested supported the above findings.

$CaCl_2$ and trace metal ions (included in the PI metal solution) are also essential to algal growth because they play important roles in various enzyme reactions. For example, in the synthesis of lipid and PUFAs, acetyl-CoA carboxylase catalyzing the initial step of fatty acid synthesis requires bivalent ions as cofactors (Singh and Ward, 1997). However, the concentrations of the ions in the medium should be at appropriate levels. High levels of trace elements inhibited the growth of *Haematococcus pluvialis* (Gong and Chen, 1997). Also, in the cultures of the marine alga *Dunaliella primolecta*, a concentration of Ca^{2+} higher than 30 μM would adversely influence the cell growth (Ohta et al., 1994). In our work, the negative effects of $CaCl_2$ and the PI solution (Table V) suggested that Ca^{2+} and trace metal ions inhibited the cell growth at the concentrations tested.

It is noticeable that both the PI and PIV metal solutions used in this work contain the trace metal ions Co^{2+}, Mn^{2+}, Zn^{2+}, and Mo^{+6} (Starr and Zeikus, 1993), and the significant effect of the PI solution was detected presumably due to the presence of Zn^{2+} and Mo^{+6}. The concentrations of Co^{2+} and Mn^{2+} in the two solutions are comparable while Zn^{2+} and Mo^{+6} in the PI solution is much higher than that in the PIV solution. When the amounts of the two solutions were respectively doubled (Tables I & II), no significant effect was detected for the PIV solution, while the PI solution showed a significant effect (Table V).

pH is another parameter influencing microalgal growth. The effects of pH on cell growth and fatty acid composition depended on the microalgal species used. For example, the diatom *Phaeodactylum tricornutum* showed good growth within the initial pH range from 6.4 to 8.4 (Yongmanitchai and Ward, 1991). While in the cultures of another diatom, *Isochrysis galbana*, the specific growth rate of the cells decreased, and the EPA percentage in the total lipids increased with decreasing pH from 8.0 to 6.0 (Molina et al., 1992). In this work, we found that the DW of *N. laevis* in high pH (9.0) was, to some extent, lower than that in low pH (7.5), while there were no apparent difference in EPA content and EPA yield at these two pH levels.

Table V shows that among the investigated factors, temperature was the most significant factor over the range tested. Cell growth was greatly inhibited when T increased from 22°C to 30°C (Tables II & V). The effects of temperature on microalgal growth might be species specific and might depend on the habit of original isolate. For example, the marine diatoms, *Phaeodactylum tricornutum* and *Isochrysis galbana* grew well at relatively low temperatures (around 20°C); the growth was greatly inhibited when temperature exceeded 25°C (Molina et al., 1992;Yongmanitchai and Ward, 1991). In contrast, the marine alga, *Crypthecodinium cohnii* could tolerate a temperature up to 31°C (Tuttle and Loeblich, 1975). Table V also shows that EPA content of *N. laevis* was higher at a lower temperature level. Similar results were reported in some algal cultures such as *Chlorella minutissima* (Seto et al., 1984), *Pythium irregulare* (Stinson et al., 1991), and *Crypthecodinium cohnii* (Jiang and Chen, 2000). The high PUFA composition at low temperature is perhaps because the cells tended to enhance the membrane fluidity by increasing the fatty acid unsaturation (Richmond, 1986). Also, the increased availability of intracellular molecular oxygen at low temperature may lead to the increase in fatty acid unsaturation because the oxygen dependent enzymes are involved in desaturation and elongation of PUFAs (Higashiyama et al., 1999; Singh and Ward, 1997).

The above results indicate that the PB design is a powerful tool for identifying factors which had significant influences on cell growth and EPA production. Nevertheless, the exact optimal values of the individual factors are still unknown but can be determined by the subsequent central composite design.

Central Composite Design

Based on the results of the PB design where the five factors had negative effects within the ranges tested, the central point and range of level for each factor were chosen (Table IV). The concentrations of nonsignificant medium components were set at their corresponding low levels in the PB design.

Table III outlines the design matrix of the central composite design (CCD) and the results obtained for DW and EPA production. The central point was repeated six times (run 27–32) to check the experimental errors. Based on the standard deviation of center point (run 27–32), the experimental errors for DW, EPA content, EPA yield, and EPA productivity were determined to be 0.18 g/L, 0.11% (of DW), 7.9 mg/L, and 0.79 mg/L/d, respectively.

Data in Table III were correlated as second-order polynomial models [Eq. (4)] by nonlinear regression using STATISTICA software. Table VI lists the estimates of coefficients and the associated *t*-values and significant levels. In this work, only the estimates with significant levels higher than 90% ($P < 0.10$) were included in the final model [Eq. (4)]. Thus, the reduced models which describe the responses as functions of the more significant variables were generated. Because the correlation of coefficient (R^2) for EPA content was rather low (< 0.65), and EPA content could be well represented by EPA yield (at constant cell

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Table VI. Estimates of coefficients of the variables in second-order polynomials [Eq. (4)] and the associated statistical tests for DW, EPA content, yield, and productivity.

		DW			EPA content			EPA yield			EPA productivity		
Coefficient	Variable	Estimate	*t*-Value	*P*-Level	Estimate	*t*-Value	*P*-Level	Estimate	*t*-Value	*P*-Level	Estimate	*t*-Value	*P*-Level
β_0	constant	7.807	31.49	**0.000**	2.88	21.77	**0.000**	226.1	32.66	**0.000**	22.47	24.79	**0.000**
β_1	[NaCl]	−0.271	−2.138	**0.055**	−0.01	−0.14	0.890	−9.287	−2.622	**0.024**	−1.009	−2.174	**0.052**
β_2	$[CaCl_2]$	−0.139	−1.100	0.295	0.08	1.14	0.279	−0.037	−0.010	0.992	−0.054	−0.116	0.910
β^3	[PI]	−0.127	−1.001	0.338	0.03	0.387	0.706	−2.465	−0.696	0.501	−0.505	−1.088	0.300
β_4	[pH]	0.354	2.788	**0.018**	−0.03	−0.412	0.688	9.241	2.609	**0.024**	1.526	3.290	**0.007**
β_5	[T]	−0.820	−6.459	**0.000**	−0.36	−5.295	**0.000**	−44.94	−12.68	**0.000**	−2.710	−5.843	**0.000**
β_{11}	$[NaCl]^2$	−0.072	−0.631	0.541	−0.08	−1.279	0.227	−9.445	−2.947	**0.013**	−0.916	−2.183	**0.052**
β_{22}	$[CaCl_2]^2$	−0.182	−1.589	0.140	0.06	1.005	0.336	−1.655	−0.516	0.616	−0.383	−0.913	0.380
β_{33}	$[PI]^2$	−0.089	−0.772	0.456	0.05	0.760	0.463	−0.194	−0.060	0.953	0.134	0.320	0.755
β_{44}	$[pH]^2$	−0.102	−0.892	0.391	0.08	1.311	0.216	2.080	0.649	0.529	−0.376	−0.896	0.390
β_{55}	$[T[^2$	−1.066	−9.289	**0.000**	0.134	2.188	**0.051**	−19.35	−6.039	**0.000**	−2.300	−5.480	**0.000**
β_{12}	$[NaCl] \cdot [CaCl_2[$	−0.191	−1.227	0.246	0.01	0.098	0.924	−5.861	−1.351	0.204	−0.397	−0.698	0.500
β_{13}	[NaCl] · [PI]	0.007	0.044	0.966	−0.003	−0.038	0.970	−0.451	−0.104	0.919	0.289	0.509	0.621
β_{14}	[NaCl] · [pH]	0.107	0.688	0.506	−0.04	−0.489	0.634	1.100	0.253	0.804	0.141	0.247	0.809
β_{15}	[NaCl] · [T]	−0.069	−0.446	0.664	−0.03	−0.339	0.741	−2.512	−0.579	0.574	0.044	0.078	0.939
β_{23}	$[CaCl_2] \cdot [PI]$	−0.070	−0.454	0.658	0.05	0.655	0.525	0.806	0.186	0.856	0.112	0.197	0.847
β_{24}	$[CaCl_2] \cdot [pH]$	0.024	0.157	0.878	0.01	0.113	0.912	1.610	0.371	0.718	−0.004	−0.008	0.994
β_{25}	$[CaCl_2] \cdot [T]$	0.373	2.401	**0.035**	−0.05	−0.670	0.516	9.135	2.105	**0.059**	1.252	2.203	**0.050**
β_{34}	[PI] · [pH]	−0.020	−0.133	0.897	−0.01	−0.113	0.912	−0.367	−0.085	0.934	−0.388	−0.683	0.508
β_{35}	[PI] · [T]	0.096	0.615	0.551	0.005	0.067	0.947	4.237	0.976	0.350	0.376	0.661	0.522
β_{45}	[pH] · [T]	0.238	1.532	0.154	−0.03	−0.324	0.752	4.541	1.046	0.318	0.612	1.077	0.304

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concentration) and EPA productivity (at constant specific growth rate), EPA content was not considered. Table VII summarizes the reduced models and the results of statistical test for DW, EPA yield, and EPA productivity. Relatively high values of R^2 were obtained for these models. *F*-test shows that the models were reliable because their significant levels were greater than 99%.

The PI metal solution was not included in the reduced models, although it was identified as an important factor by the PB design. This was due to the different concentration ranges used in the two experiments. In the CCD experiments, the concentrations of the PI solution tested (4.5 to 13.5 mL/L) showed no significant effects on DW and EPA production.

To determine the optimal levels of the variables, the three-dimension surface plots were constructed according to the equations in Table VII. Each response was plotted as the function of NaCl, pH (T and $CaCl_2$ were kept constant) and the function of T, $CaCl_2$ (NaCl and pH were kept constant), respectively. Because only T and $CaCl_2$ interacted in the models, while NaCl and pH had no interaction with T and $CaCl_2$, the optimal data of the two independent variables in the surface plots (e.g., NaCl and pH) were not influenced by the chosen (fixed) levels of the other two variables (e.g., T and $CaCl_2$), and vice versa.

Figure 1A shows that NaCl and pH linearly affect DW. In the range tested, DW increased with decreasing NaCl and increasing pH; the maximum DW was obtained at the −2 level of NaCl and the +2 level of pH. Figure 1B shows DW as the function of T and $CaCl_2$, a quadratic effect of T and an interactive effect of T and $CaCl_2$ on the response were observed. DW reached the maximum value at the T level between −0.5 to −1.0 and the −2 level of $CaCl_2$.

The surface plots of EPA yield are presented in Figure 2. As shown in Figure 2A, NaCl had linear and quadratic effects, while pH had only a linear effect on the response. The maximum EPA yield was obtained at about the −0.5 level of NaCl, and the +2 level of pH. Figure 2B shows that a decrease in $CaCl_2$ led to a slight increase in EPA yield while the optimal T for EPA yield was influenced by the chosen level of $CaCl_2$. According to the plot, it was found that the optimal levels of T and $CaCl_2$ for EPA yield were −1.5 and −2, respectively.

Figure 3A and B show the surface plots of EPA productivity as the function of NaCl, pH, and T, $CaCl_2$, respectively. The patterns of the plots were similar to the corresponding plots for EPA yield (Fig. 2) because the two re-

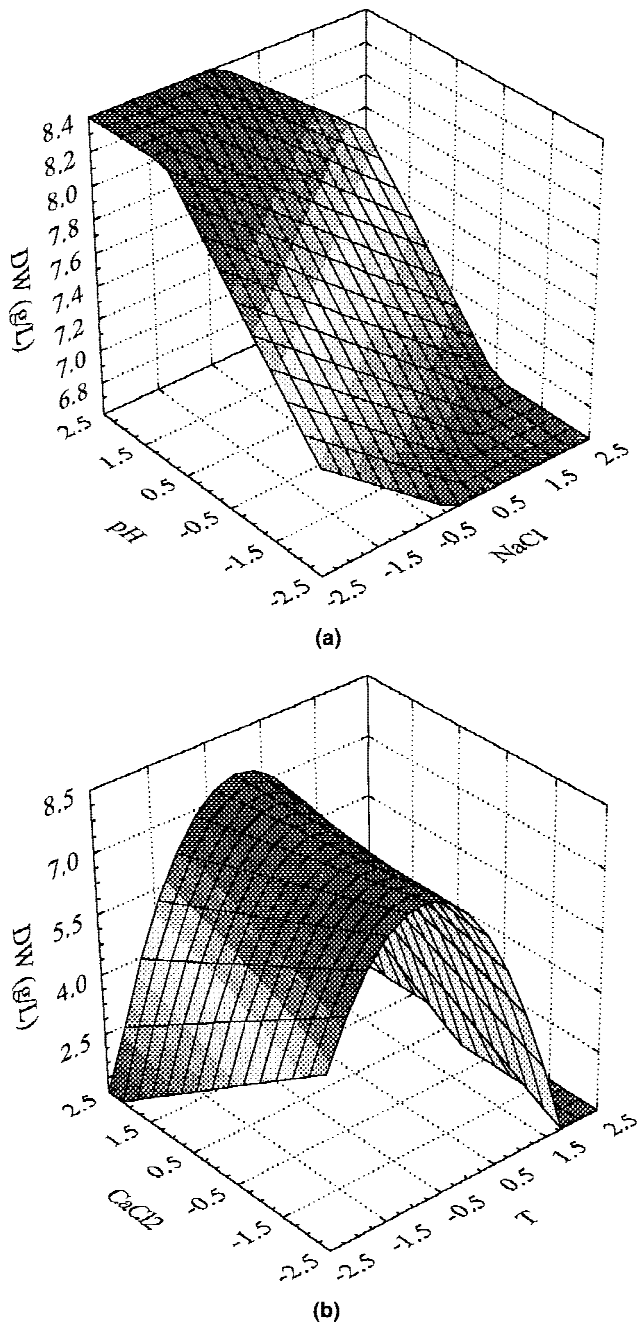

Figure 1. Three-dimensional surface plots of DW as the function of NaCl and pH in coded unit [T and $CaCl_2$ were kept at central point (A)], and as the function of T and $CaCl_2$ in coded unit [NaCl and pH were kept at central point (B)].

Table VII. Reduced models for DW, EPA yield, and EPA productivity as the functions of independent variables.[a,b,c]

Responses	Unit	Model	R^2	*F*-Value
DW	g/L	$7.451 - 0.271[NaCl] + 0.354[pH] - 0.820[T] - 1.036[T]^2 + 0.373[CaCl_2][T]$	0.867	33.85
EPA yield	mg/L	$226.3 - 9.3[NaCl] + 9.2[pH] - 44.9[T] - 9.5[NaCl]^2 - 19.4[T]^2 + 9.135[CaCl_2][T]$	0.930	55.42
EPA productivity	mg/L/d	$21.93 - 1.0[NaCl] + 1.53[pH] - 2.71[T] - 0.87[NaCl]^2 + 1.25[CaCl_2][T]$	0.838	21.55

[a]Only coefficients (Table VI) in second-order polynomial [Eq. (4)] with $p < 0.10$ were included.
[b][NaCl], [pH], [T], [$CaCl_2$] represent the coded levels of NaCl, pH, temperature, and $CaCl_2$, respectively.
[c]For all the models, a significant level >99.9% was obtained.

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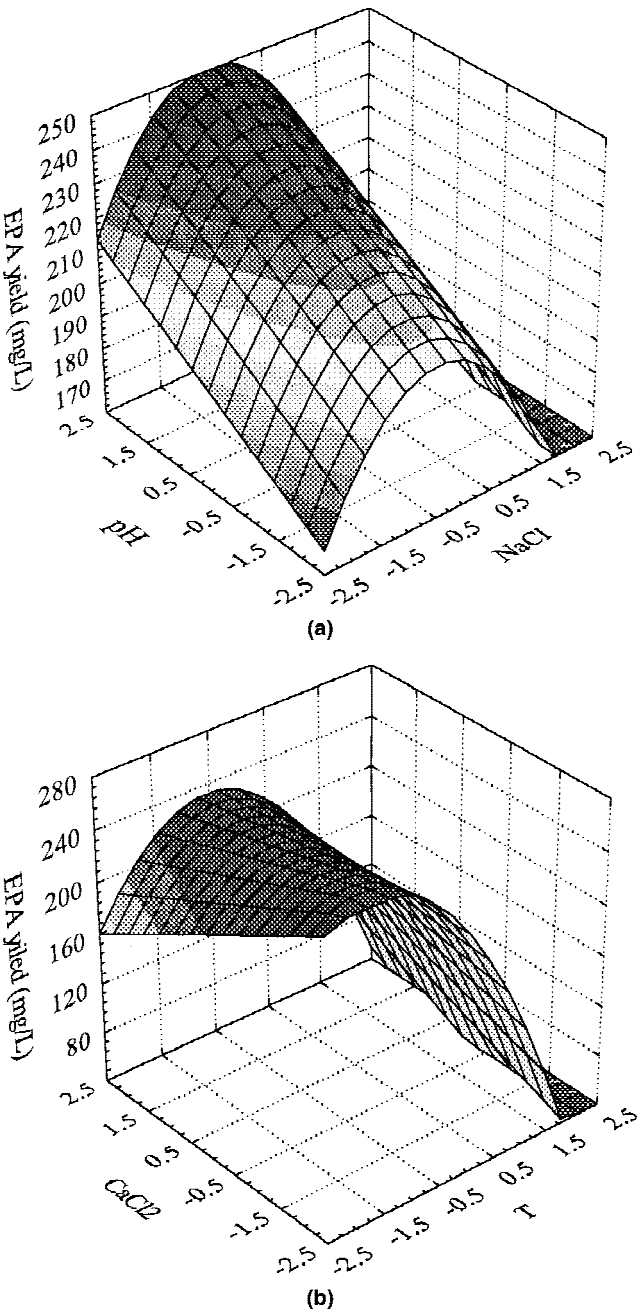

Figure 2. Three-dimensional surface plots of EPA yield as the function of NaCl and pH in coded unit [T and $CaCl_2$ were kept at central point (A)], and as the function of T and $CaCl_2$ in coded unit [NaCl and pH were kept at central point (B)].

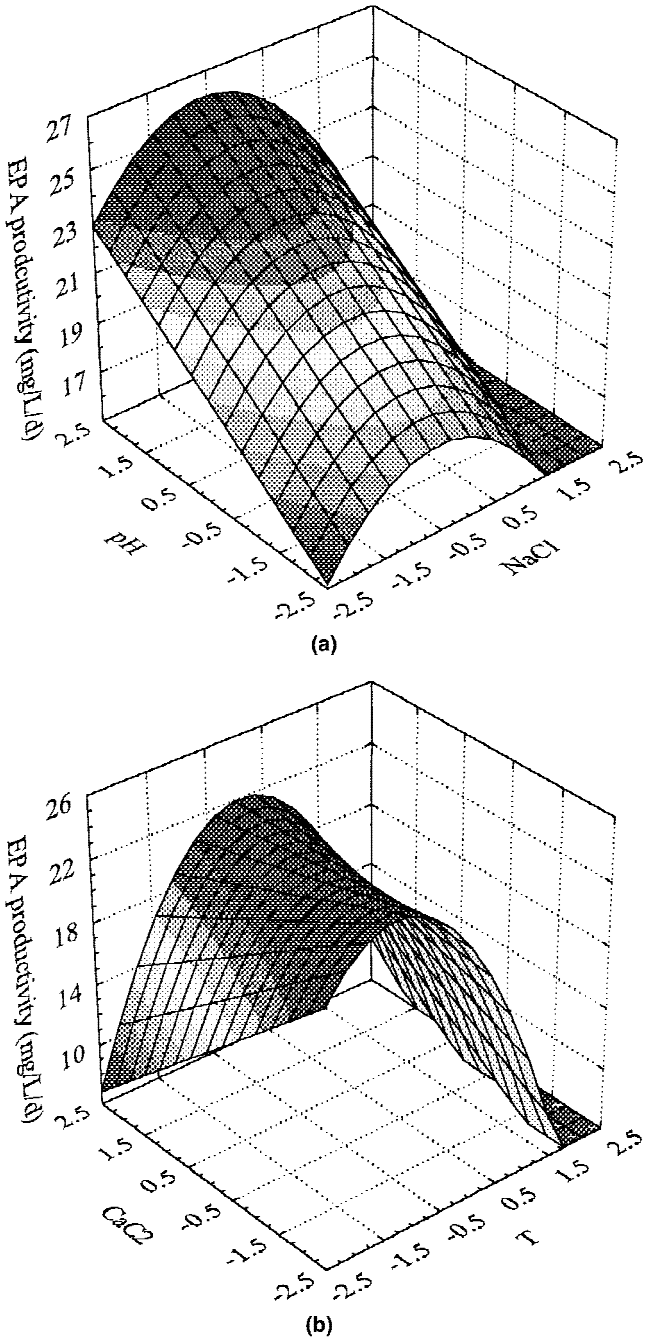

Figure 3. Three-dimensional surface plots of EPA productivity as the function of NaCl and pH in coded unit [T and $CaCl_2$ were kept at central point (A)], and as the function of T and $CaCl_2$ in coded unit [NaCl and pH were kept at central point (B)].

sponses had the same terms in the reduced models (Table VII). According to Figure 3, the optimal values (in coded level) of variables were determined as NaCl: ≈ 0.5, pH: +2, $CaCl_2$: −2, and T: −0.5 to −1.5.

Through these surface plots, an estimation of the optimal level for each variable could be obtained. We also used numerical technique with the software MAPLE (V4.0, Waterloo Maple Inc., Waterloo, Canada, 1996) to obtain the accurate optimal values of the variables. The results agreed with those obtained by surface response analyses. Table VIII summarizes the optimal levels of the variables (in both coded and real values) for DW and EPA production. The predicted data of responses in each setting of variables are also presented. In the next step, experiments were carried out to verify these optimized culture conditions.

Verification

Verification experiments were conducted under two different culture conditions. One was under the optimal condi-

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Table VIII. Optimal values of NaCl, pH, temperature (T), and $CaCl_2$ (in coded and real values) for DW, EPA yield, and EPA productivity with corresponding predicted data.

Variable	NaCl		pH		T		$CaCl_2$		
Parameter	Coded	Real	Coded	Real	Coded	Real	Coded	Real	Predicted data
DW	−2	g g/L	2	8.5	−0.76	19.80°C	−2	0.10 g/L	9.29 g/L
EPA yield	−0.49	14 g/L	2	8.5	−1.62	17.13°C	−2	0.10 g/L	298.4 mg/L
EPA productivity	−0.57	13.7 g/L	2	8.5	−1.16	18.62°C	−2	0.10 g/L	28.29 mg/L/d

tions for DW (experiment A: NaCl = 8g/L, pH = 8.5, $CaCl_2$ = 0.10g/L, T = 20°C). Because the optimal culture conditions for EPA yield and EPA productivity were very similar (Table VIII), the other experiment was done under the suboptimal conditions of variables for EPA yield and productivity (experiment B: NaCl = 14g/L, pH = 8.5, $CaCl_2$ = 0.10g/L, T = 18°C). Although NaCl concentration and temperature in experiment B deviated slightly from their optimal values (Table VIII), it was found that the predicted data of EPA yield and EPA productivity in experiment B were only slightly lower (< 1%) than the optimal values (Tables VIII & X).

Table IX shows that the fatty acid profiles of *N. laevis* in the two experiments. The compositions of fatty acids were almost identical as previous results (Wen and Chen, 2000). As shown in Table X, there was a good agreement (less than 8% error) between the experimental data and the corresponding predictions. A high DW of ≈ 9 g/L, an EPA yield of ≈ 280 mg/L and an EPA productivity of ≈ 28 mg/L/d were obtained in the optimized culture conditions. In our previous studies, DW, EPA yield, and EPA productivity were only 6.4 g/L, 180 mg/L, and 20 mg/L/d, respectively. It should be noted that as temperature was the significant factor for EPA content ($p < 0.001$, Table VI). The EPA content of 31 mg/g DW (280 mg/L divided by 9 g/L) obtained at optimal EPA yield condition (T = ≈ 18°C) was considered significantly different from that of previous studies (i.e., 28 mg/g DW of EPA content, 180 mg/L over 6.4 g/L), in which T was controlled at 25°C. Thus, the high volumetric EPA production achieved in this work was attributed to both higher cell concentration and EPA content. However, if compared the corresponding data of this work to those of previous study, it was found that EPA content increased about 10% (from 28 mg/g DW to 31 mg/g DW) while cell concentration increased about 40% (from 6.4 g/L to 9g/L). The contribution of EPA content to the improvement of volumetric EPA production is therefore, lower than that of cell concentration.

Table IX. Fatty acid composition (% total fatty acid) of N. laevis at optimal levels of NaCl, $CaCl_2$, pH, and temperature for DW (experiment A) and for EPA yield and EPA productivity (experiment B).[1]

Fatty acids	Experiment A	Experiment B
14:0	9.81 ± 0.58	10.51 ± 0.29
14:1	1.25 ± 0.09	0.96 ± 0.04
16:0	27.46 ± 0.70	26.07 ± 0.80
16:1	39.81 ± 0.57	40.83 ± 0.98
18:0	1.36 ± 0.06	1.07 ± 0.15
18:1	2.42 ± 0.37	2.19 ± 0.08
18:2	2.72 ± 0.18	2.44 ± 0.19
18:3 (ω − 6)	0.76 ± 0.15	0.70 ± 0.02
18:3 (ω − 3)	0.74 ± 0.14	0.64 ± 0.09
20:4 (ω − 6)	2.96 ± 0.12	3.19 ± 0.04
20:5	10.67 ± 0.54	11.38 ± 0.51
TFA content (%DW)	28.26 ± 1.20	29.67 ± 0.82
EPA content (%DW)	3.01 ± 0.14	3.38 ± 0.12

[a]Data are expressed as mean ± standard deviation of five replicates.

Table VIII also shows that the optimal values of NaCl and $CaCl_2$ for DW were located on the "border" of the experimental range investigated (−2 to +2). An extrapolation of the equations (in Table VII) beyond this variable range would result in higher value of DW. To test this hypothesis, the cells were cultured in the medium in which the concentrations of NaCl and $CaCl_2$ were respectively beyond the ranges tested in the CCD (i.e., NaCl: 4 g/L, −3 level; $CaCl_2$: 0.05 g/L, −3 level), while the other variables were kept at their corresponding optimal values (for DW). It was found that there were no further improvements in DW at these settings compared with the optimal culture conditions shown in table VIII (data not shown). Similarly, when $CaCl_2$ was set at 0.05 g/L (−3 level) and other variables were kept at their optimal values (for EPA production), the results also indicated no improvement in EPA yield and productivity. As for pH, the optimal value (+2 level, pH 8.5) was also located on the "border" of the tested range. However, the screening experiment had shown that high pH (9.0) caused negative effect on cell growth compared with low pH (7.5) (Table V). Therefore, pH 8.5 was considered to be the best value for the cultures of *N. laevis*. Based on the above results, we confirmed the optimal values of NaCl, $CaCl_2$, and pH that were obtained by the CCD experimental designs.

The above results showed that a lower NaCl concentration was optimal for cell growth, while a higher NaCl concentration favored EPA production. Similar trends were reported in the cultures of the green alga *Chlorella minutissima* (Seto et al., 1984). The optimal $CaCl_2$ concentration, pH, and temperature for growth and EPA production of *N. laevis* are almost identical. The results agreed with previous

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Table X. Comparison of experimental and predicted data of DW, EPA yield, and EPA productivity at optimal culture conditions (experiment A, and experiment B).[a]

Experiment		DW	EPA yield	EPA productivity
A	Experimental	8.65 ± 0.20	260.5 ± 7.53	26.05 ± 0.75
	Predicted	9.28	258.9	25.99
	Error (%)	−7.28	+0.61	+0.23
B	Experimental	8.38 ± 0.18	283.2 ± 6.93	28.32 ± 0.88
	Predicted	8.54	296.7	28.22
	Error (%)	−1.91	−4.77	+0.35

[a]Experimental data are expressed as mean ± standard deviation of five replicates.

reports on other microalgae using conventional method (Molina et al., 1992;Yongmanitchai and Ward 1991). However, compared with the conventional one-at-a-time approach, the presented approach contributed to the knowledge for optimizing EPA production in the following aspects. (1) It improved the accuracy in parameter assessment. For example, in the culture of *C. minutissima* (Seto et al., 1984), although the trends of cell growth and EPA yield with NaCl concentration is known, it was difficult to determine the precise optimal NaCl concentration from the experimental results. In the case of *P. tricornutum* (Yongmanitchai and Ward, 1991), the pH for good growth was found in the range 6.4–8.4; the exact pH value is still unknown. For the alga *I. galbana* (Molina et al., 1992), the EPA content was higher over a wide range of temperature (15–25°C); the exact optimal temperature could not be predicted as the interval between two actually investigated temperatures was rather large (5°C). In this study, as second-polynomial equations were established by fitting the experimental data (Table VII), the exact optimal values of the parameters could be interpolated, i.e., predicting the response at factor levels between those discretely used in the experiment. (2) The present approach proved more efficient than the one-at-a-time method. In the culture of *P. tricornutum*, Yongmanitchai and Ward (1991) investigated factors affecting cell growth and EPA production, including nitrogen sources, phosphate, NaCl, vitamin, silicate, CO_2, oleic acid, temperature, pH, and inoculum size, which was complicated and time-consuming. In this study, relatively few experiments were necessary to accomplish the entire work including 13 variables-screening, 5 variables-optimization, and 2 conditions-verification. (3) The interactions between $CaCl_2$ and temperature were found to influence both the growth and EPA production of *N. laevis*. Such interaction could not be determined if the one-at-a-time method was used.

In short, we aimed at optimizing EPA production by *N. laevis* in this work, and statistically based experimental designs served as efficient tools in this optimization process. As the effect of a factor was estimated at several levels of other factors, the conclusions obtained were valid over a range of experimental conditions. However, the underlying mechanisms by which these factors affect EPA production are still not very clear. Further research is required to elucidate these mechanisms.

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